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The British Motorcycle Industry at a Crossroads

By 1975, the collapse of the British motorcycle makers was nearly complete. In the face of competition from Honda and other Japanese rivals, British production had fallen from 130,000 bikes in 1960 to 20,000 in 1974 and seemed unlikely to recover in 1975.¹ A single British company, Norton Villiers Triumph (NVT), remained in operation. The company, a recent combination of often-reorganized Norton Villiers and failing BSA/ Triumph, lost £5.1 million in 1974 on sales of roughly £16 million.² In March, 1975, Britain's Secretary of State for Industry asked consultants from the Boston Consulting Group (BCG) to identify and evaluate strategic alternatives for the last British motorcycle producer and its suppliers. This case summarizes what the consultants discovered as they sized up the global motorcycle industry and Britain's position in it.

Markets and Products

British motorcycles were sold mostly in North America and Europe. In total, consumers bought more than 1.4 million motorcycles in those regions in 1974.³ Buyers in these areas used motorcycles mostly for recreation, not for basic transportation, and with the introduction of new recreational products such as off-road bikes and so-called superbikes, unit sales had risen nearly 16% per year since 1968. Demand appeared to be reaching saturation, however, and the BCG consultants forecast a 4-5% annual growth rate for the coming years.⁴

Nine out of ten motorcycle buyers in the U.S., the world's largest motorcycle market, were men, and more than half were younger than 25. The typical buyer shopped for eight weeks before purchasing and gathered information from many sources, including other motorcycle owners (consulted by 85% of buyers), dealers (82%), cycling magazines (69%), manufacturers' literature (52%), competition results (39%), and test drives (38%).⁵ **Exhibit 1** shows the product attributes that buyers considered most important.

Products ranged from small bikes designed for economical commuting to luxurious superbikes equipped for cross-country cruising to rugged off-road cycles with studded tires. Bikes were often classified by engine displacement, which corresponded roughly to size and engine power. Common categories included learner's bikes (<125 cc displacement), intermediate bikes (125-349 cc), medium bikes (350-449 cc), big bikes (450-749 cc), and superbikes (≥750 cc).⁶ **Exhibit 2** breaks down the U.S. market by displacement class and shows the recent growth rate in each class. First-time buyers

Professor Jan W. Rivkin prepared this case from published sources, especially the report "Strategy Alternatives for the British Motorcycle Industry" by the Boston Consulting Group Limited, published in 1975 by Her Majesty's Stationery Office. The case is designed to be taught with the earlier cases "Honda (A)" (384-049) and "Honda (B)" (384-050). HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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typically started with a small-displacement motorcycle, then traded up over time, usually within a year. When trading up, roughly a third of buyers stayed with the same brand.⁷

Prices typically rose in line with displacement, as shown in **Exhibit 3** for 1969 and 1975. Within a displacement class, however, competitors did set different prices from one another. BCG's consultants felt that demand for a particular model would neither rise nor fall much as long as the model's price was within 10% of the typical price for its displacement class. Premiums greater than 10% would cut demand for the model significantly, and premiums above 40% would wipe out essentially all demand.⁸

Competitors

The dominant dynamic in the global motorcycle industry during the 1960s and early 1970s was the rise of Japanese competitors, led by Honda. The story of Honda's overseas expansion is laid out in related case studies.⁹ Honda's Japanese rivals – Yamaha, Kawasaki, and Suzuki – differed from Honda in several ways, but for simplicity in this case, we will take Honda as representative of the Japanese competitors. **Exhibit 4** shows the shifts in sales and market share in the United States between 1968 and 1974. **Exhibit 5** shows the number of models offered by each competitor in 1968 and 1974, by displacement class. In its globalization efforts, Honda had succeeded first in the market for small motorcycles, but then expanded into larger displacement classes.

In the United States, Honda and the other Japanese rivals faced Harley-Davidson, BMW, and NVT. Harley-Davidson maintained a loyal following in the U.S., both because it was the last U.S.-made option for patriotic customers and because its big, tough bikes closely matched a big, tough set of leather-clad customers. Observers noted that Harley was among the few brand names that customers commonly tattooed on their bodies. In 1975, Harley was petitioning the U.S. International Trade Commission to raise tariffs on motorcycles in order to protect U.S. employment. BMW produced motorcycles only in the 600-900 cc range and made 25,000 machines per year worldwide. Expansion plans called for an output of 45,000 machines per year by 1980. All BMW models were variants of a single engine and frame design, a design which led *Motorcycle Magazine* to name the BMW 750 the best superbike in the world in 1973.¹⁰ With a reputation for very high quality, BMW charged a premium of 30-40% over rival bikes of equal displacement.

In the past, NVT and its predecessors had produced a full range of motorcycles. Individual models had been withdrawn, however, as each became unprofitable. Since the early 1960s, lack of model profitability had prompted the British to exit the 175, 250, 350, 500, and 650 cc classes.¹¹ In 1975, all of NVT's remaining models had displacements of 740 cc or greater. British makers had introduced no genuinely new models since Triumph's Trident in 1968 though many models had been "uprated" to higher displacements. **Exhibit 6** shows British output over time.

British motorcycles had been popular around the world at least since Marlon Brando rode astride a Triumph in the 1954 motorcycle-gang film *The Wild One* (a film whose violent themes caused it to be banned in Britain for more than a decade). Triumphs continued to make regular appearances in Hollywood films in the 1970s. British bikes had a traditional reputation for premium quality finish, elegant styling, superior handling, and high performance (e.g., quick acceleration and high top speeds). By 1975, however, new Japanese machines such as the Kawasaki Z-1 easily topped the British bikes in performance tests, and technological changes had improved their handling characteristics. New Japanese models incorporated features such as electric starters, disc brakes, and five-speed transmissions, features which British makers struggled to add to existing models. British motorcycles were also known for their mechanical failures, particularly oil leaks. Warranty claims

were resisted by dealers, and spare parts were hard to come by.¹² Observers joked that British bikes were particularly popular among riders who enjoyed fixing their own machines.

British models tended to be priced at or slightly above the prices of Japanese models with comparable displacements, but pricing patterns varied widely over time. In 1973, for instance, Triumph's Bonneville 750 model was priced 12% below the Honda 750 in the U.S.; indeed, 1973 was the only year since 1968 that Triumph had not lost market share in the U.S. In 1975, in an effort to regain profitability, NVT priced its models nearly \$400 above comparable Japanese models.¹³ A typical NVT bike had a retail price of \$2,495 while a comparable Honda 750 was priced at \$2,112.

Design and Production

Exhibit 7 lays out the costs incurred to bring a typical British motorcycle to market in the United States. More than two-thirds of the manufacturer's costs were incurred before the machine left the factory. Design of a motorcycle preceded production, which involved the procurement of raw materials and components, the production of engine parts, the production of frame parts, and assembly. Honda and NVT took very different approaches to each of these steps.

Design. At Honda, marketing personnel performed a commercial evaluation of any new product or feature and production engineers assessed its costs before design of the product or feature began. Design followed a precise program, with regular checkpoints and reassessments and with special attention to a design's implications for unit costs. Honda's subsidiary, the Honda R&D Company, employed 1,300 people, or 0.75 people per \$1 million of corporate motorcycle sales. Each Honda R&D employee, it was estimated, earned \$17,000 per year in 1975, and personnel accounted for roughly half of total R&D expenses. Honda claimed that its design personnel could take an idea from conception to production in 18 months.¹⁴

NVT's engineering staff numbered roughly 100, or 2.7 people per \$1 million of corporate sales. The staff tended to focus on "pure" design considerations and would use non-standard parts and unusual fittings if necessary to improve performance, handling, or styling. The joints between crankcase parts and cylinder block, for instance, met in a "T" shape, consistent with classic British styling but nearly impossible to seal tightly. Projects were controlled informally, with virtually no paperwork. The engineering staff estimated that a minimum of 18 months was required to move an idea from conception to a complete set of drawings and 18 more months were needed to tool up for production.¹⁵

Production of engine and frame parts. Honda carried out its global production of nearly 2 million bikes per year in four very modern plants in Japan. Plants were specialized: one plant, for instance, made only engines. In the formation and machining of engine and frame parts, Honda made extensive use of relatively automated die-casting techniques and machine tools. BCG's consultants estimated that a company unfamiliar with automated technology, such as NVT, might have to invest \$48 million in plant and equipment with a 10-year depreciable life in order to adopt the technology. The technology would reduce the total cost per unit of such a company – fixed costs per unit plus variable costs per unit – once its production of similar components exceeded 50,000 units per year. (Below this volume, older manual technology would lead to a lower total cost per unit than the automated systems.) Because Honda grouped its many models into families that used similar components, it was not unusual for Honda to require 100,000 units of a particular component each year, worldwide. Indeed, the \$48 million investment mentioned above would provide enough equipment to produce the various engine and frame parts for roughly 100,000 motorcycles and not many more. Beyond the cost savings attained by employing automated equipment at large scale,

Honda enjoyed the benefits of extensive production experience. BCG estimated that accumulated experience might have reduced Honda's unit costs of engine and frame part production by 20% or even more.¹⁶

Honda's machine tools were highly specialized, and many were custom-made by its 1,400-person Honda Engineering subsidiary. Honda Engineering delivered machine tools less than six months after an order was placed. Delivery times of 18 months were common on the open market.¹⁷

NVT produced its motorcycles in three factories in Wolverhampton, Small Heath, and Meriden, all in the Midlands section of England. Factory buildings dated from the First World War period. Small Heath was multi-story, and Wolverhampton, with many separate buildings, was described by BCG consultants as "rather confused."¹⁸ Within the buildings, general purpose, labor intensive machines were used to manufacture engine and frame parts. At Wolverhampton, it was estimated that 80 percent of machine tools were more than 15 years old. It was not unusual for parts for a single motorcycle to come from machine tools of quite different vintages and tolerances. The costs associated with parts production were essentially all variable, since the fixed costs associated with plant and equipment were nearly completely depreciated. Though BCG consultants detected a strong, downward sloping experience curve for Japanese manufacturers, they found no association between unit costs and accumulated production experience for NVT.¹⁹

Assembly. Assembly of finished product was the least automated and most labor-intensive stage of Honda's production process. Most of the 16,400 employees at Honda's four factories were involved in assembly. (The average factory laborer at Honda earned \$8,500 in 1975.) Powered conveyors and sophisticated factory control techniques allowed Honda to keep assembly cycle times down to a bare minimum.

In NVT's factories, most employees manned the machine tools that produced engine and frame parts, but a significant minority were involved in assembly. (The average factory laborer at NVT earned \$5,800 in 1975.) It was estimated that the average Honda assembler could put together 2.3 motorcycles in the time it took the average NVT assembler to make one. The difference arose in part because Honda made many simple, small-displacement motorcycles with few parts. For bikes of equal displacement, the ratio was perhaps 1.5 to 1.²⁰

A series of layoffs, mergers, and reorganizations had left labor relations strained and worker morale low at NVT's three factories. Morale was somewhat better at Meriden than elsewhere. When Norton Villiers and BSA / Triumph merged in 1973, management tried to close Meriden, but workers occupied the factory in protest. The sit-in led to the establishment of the Meriden Workers' Cooperative which, with an infusion of cash from the British government, purchased most of the assets at Meriden. Thus Meriden was formally owned by the workers, though in practice it operated as part of NVT.²¹

Procurement of other components. Honda and NVT alike purchased significant components from outside suppliers. Honda worked closely with its suppliers to reduce working capital and to co-locate production, in some cases even allowing suppliers to set up operations within Honda plants. Honda also suggested new production methods and technologies to suppliers and worked with them to integrate R&D plans. Suppliers tended to invest in automated production techniques, and they appeared to enjoy benefits of automation, scale, and experience similar in proportion to those enjoyed by Honda itself in its production of engine and frame parts. The typical Honda supplier earned a return on assets in the 6-10% range.²²

NVT suppliers tended to use production techniques comparable to those of NVT itself, not the latest, high-volume production techniques. Exceptions arose when NVT procured parts from large auto industry suppliers such as Dunlop. Such suppliers considered NVT a very small customer.²³

Selling and Distribution

The distribution subsidiaries of Honda, NVT, and other competitors sold motorcycles to independent dealerships, which in turn sold the machines to riders. The subsidiaries handled physical distribution, deployed advertising and promotions to pull consumers into dealerships, and gave dealers incentives and support to push bikes through the channel.

Physical distribution was straightforward. The companies bought freight services and packing materials in competitive markets on equal terms. When shipping abroad, they also paid duties. In the United States, duties on a motorcycle totaled about 3% of the cycle's retail price.

Advertising and promotion expenditures focused on television spots and on ads in consumer and trade magazines. In the United States in 1975, Honda and NVT had consumer advertising and promotion budgets of \$10 million and \$800,000, respectively.

Dealer relations included the most complicated aspects of the selling and distribution task. In the United States, most dealerships were small businesses with annual sales of less than \$200,000 and single-digit net margins. The number of American dealers had swelled from roughly 3,000 in 1960 to 10,000 in 1974.²⁴ The distribution subsidiary of each motorcycle maker fielded a sales force that reached out to the numerous dealers and, in many cases, convinced dealers to offer its brand of bike exclusively. On an on-going basis, the motorcycle makers supported dealers by distributing spare parts, paying for service that was performed under warranty, helping with product display and market planning, training sales staff, and providing financial assistance if needed.

In 1975, Honda sold its motorcycles in the United States through 1,974 dealers, 84% of which sold Honda bikes exclusively. Honda was known among dealers for extensive sales support, for generous financial assistance that could be used to modernize showrooms, and for a computerized system that made 14,000 spare parts for its diverse product line readily available. In its early days in the U.S., Honda had also offered dealers an unusually fat markup between the wholesale price and the suggested retail price. By 1975, however, because of the rapid turnover of Honda-branded inventory, dealers were willing to accept a dollar markup on each Honda bike that was as much as \$120 lower than the markup on each British cycle.²⁵

NVT's U.S. distribution network was in flux in 1975. Prior to the merger, in 1972, BSA / Triumph had 850 dealers in the U.S. and Norton had 750. By 1975, Triumph was down to 408 dealers, 7% of which sold Triumph machines exclusively; the number of Norton dealers was not reported by the BCG consultants. NVT hoped to whittle the total number of U.S. dealers down to 400 eventually. NVT offered considerably less support to its dealers than did Honda or other Japanese competitors.²⁶

The BCG consultants argued that unit costs associated with selling and distribution should fall as a company's overall sales volume in a market, its volume per model, and its volume per dealer rose, and unit costs should rise as a company offered its dealers more extensive support. **Exhibit 8** gives data on a number of these factors. The consultants also noted a strong pattern across companies: for virtually all companies, including Honda and NVT, total selling and distribution costs were 20% of overall retail sales.²⁷

Overall Performance. NVT's 1974 loss of £5.1 million, or \$12.1 million, on sales of £16 million, or \$37 million, overstated the company's desperation slightly. The loss included a substantial charge associated with the merger of Norton Villiers and BSA / Triumph. Even excluding the charge, however, NVT lost money in 1974. The company felt that because of price increases it had made between 1974 and 1975, bikes sold in the U.S. during 1975 would make a small but positive contribution to covering the company's modest overhead expenses. This positive contribution is reflected in **Exhibit 7**.

Worldwide, Honda had motorcycle revenue of \$1.7 billion in 1974. Honda's profit margins were difficult to determine precisely because the company participated in other businesses, especially the auto market, where it was in the midst of an expensive effort to expand around the world. Observers speculated that return on sales in the motorcycle business was in the single digits, with substantially higher margins possible for large-displacement bikes.

Strategic Alternatives

Having analyzed the motorcycle industry and Britain's place in it, the BCG consultants began to formulate strategic alternatives for NVT. They first laid out some broad objectives. They then identified a set of options that might meet the objectives. All but three options seemed highly unlikely to work. For each of those three, the BCG team made a detailed projection of volumes, prices, costs, and required investments. From each projection came a forecast of operating results, financial returns, employment levels, and the industry's contribution to the U.K. balance of payments. The team also examined the sensitivity of each forecast to competitive price cuts and to errors in the volume projection, and it assessed the implementation risks associated with each option. The consultants stopped short of a final recommendation, however. The choice among the viable options, the consultants noted, was likely to involve "non-commercial considerations outside the scope of the study."²⁸

Exhibit 1 Factors Influencing U.S. Motorcycle Buyers

Product Attribute	Percent of Buyers Rating Attribute Important or Very Important
Quality of workmanship	94.6
Availability of parts	91.6
Handling / performance	88.9
Power / acceleration	85.9
Styling / appearance	84.6
Recommendation of owners / friends	81.3
Dealer's reputation	80.4
Dealer's service	76.9
Resale value	74.8
Warranty coverage	72.1
Economy of operation	63.9
Test drive	56.3
Owned same make before	34.3

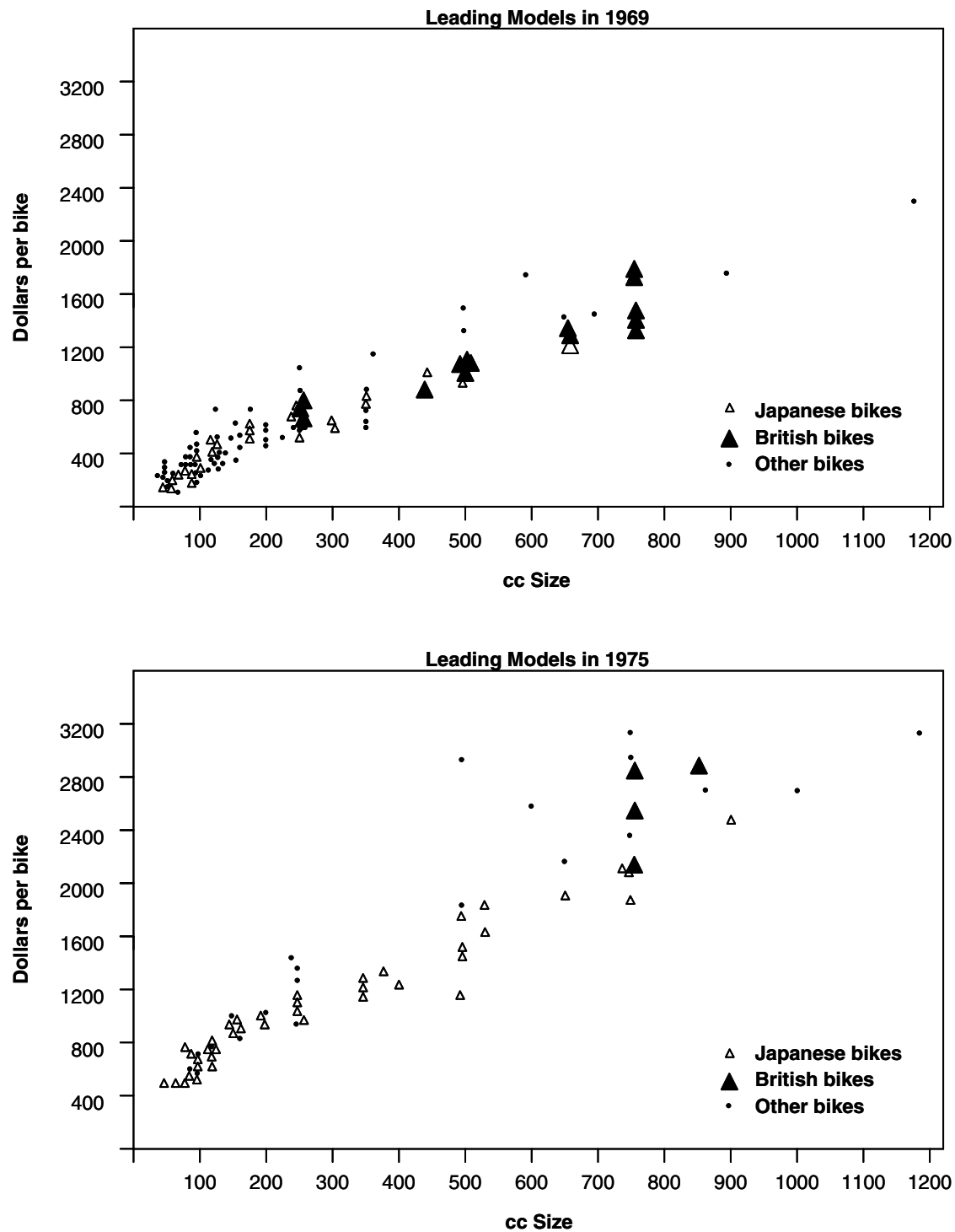
Source: AHF Marketing Research, Inc., "The Market for New Motorcycles," 1973, reported in D. Purkayastha and R.D. Buzzell, "Note on the Motorcycle Industry – 1975," HBS case 578-210, p. 4.

Exhibit 2 Size and Growth of U.S. Motorcycle Market, by Displacement Class

Class	Units sold in 1974 (000s)	Retail value in 1974 (\$ mm)	Average retail price in 1974 (\$)	Unit CAGR, 1968-74 (%)
<125 cc	276	81	293	2.2
125-349 cc	382	339	887	12.8
350-449 cc	200	230	1,150	25.8
450-749 cc	173	186	1,076	16.7
750 cc and above	128	288	2,253	46.7
Total	1,159	1,124	970	15.2

Source: Motorcycle Industry Council, reported in BCG, "Strategy Alternatives for the British Motorcycle Industry," Appendix pp. 7-8, 18-20.

Exhibit 3 Price / Displacement Relationship



Source: NADA Motorcycle Appraisal Guide, reported in BCG, "Strategy Alternatives for the British Motorcycle Industry," Appendix pp. 51, 53.

Exhibit 4 Sales and Market Shares in the United States

Competitor	<u>Units sold (000s)</u>		<u>Retail value (\$ mm)</u>		<u>Unit share (%)</u>		<u>Average price (\$)</u>		<u>Number of models</u>		<u>Units per model (000s)</u>	
	1968	1974	1968	1974	1968	1974	1968	1974	1968	1975	1968	1974/75
Honda	161.1	433.8	98.4	477.7	37.5	37.4	611	1,101	21	26	7.7	16.7
Yamaha	58.2	205.1	31.2	215.4	13.5	17.7	536	1,050	22	29	2.6	7.1
Kawasaki	19.9	129.6	8.9	147.0	4.6	11.2	446	1,134	15	23	1.3	5.6
Suzuki	23.0	108.5	8.7	98.1	5.3	9.4	379	904	4	24	5.7	4.5
Harley-Davidson	26.3	57.9	27.8	111.8	6.1	5.0	1,056	1,930	6	15	4.4	3.9
BMW	2.4	9.0	3.1	25.8	0.6	0.8	1,305	2,854	3	4	0.8	2.3
NVT	27.8	13.9	34.5	26.3	6.5	1.2	1,242	1,893	20	17	1.4	0.8
Others	111.3	201.1		21.9	25.9	17.4		109				
Total	430.0	1,159.0		1,124.0				970				

Source: R.L. Polk and BCG estimates, reported in BCG, "Strategy Alternatives for the British Motorcycle Industry," Appendix p. 23. NADA Motorcycle Appraisal Guide, reported in D. Purkayastha and R.D. Buzzell, "Note on the Motorcycle Industry – 1975," HBS case 578-210, p. 21. Casewriter calculations.

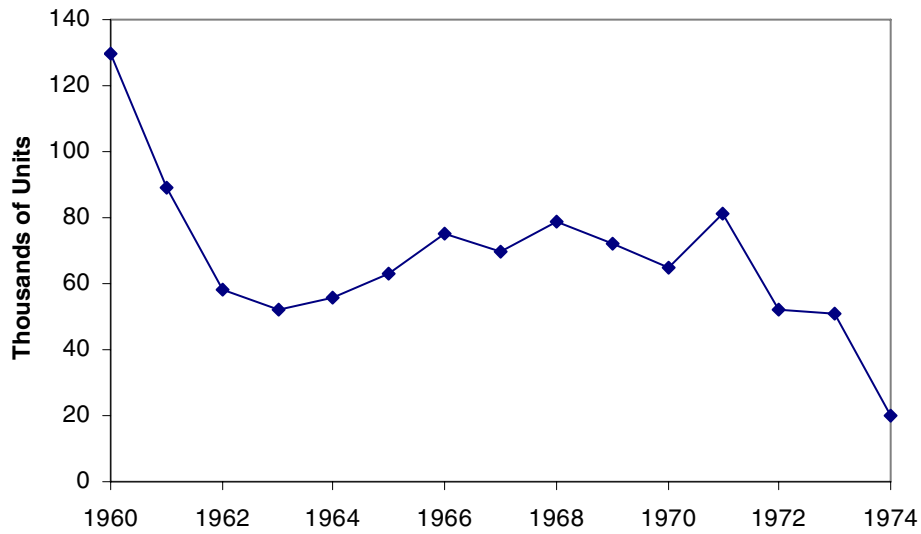
Note: Differences in average prices across competitors reflect differences in displacement-class mixes, not just discounts given or premiums commanded within a displacement class. The average price shown here for NVT differs from the price shown in Exhibit 7 primarily because of price adjustments and inflation between 1974 and 1975.

Exhibit 5 Models Offered in Each Displacement Class

Competitor	<u>Under 125 cc</u>		<u>125-349 cc</u>		<u>350-449 cc</u>		<u>450-749 cc</u>		<u>750 cc and above</u>	
	1968	1975	1968	1975	1968	1975	1968	1975	1968	1975
Honda	10	14	9	6	2	3	0	3	0	0
Yamaha	11	12	11	11	0	4	0	2	0	0
Kawasaki	3	10	9	6	0	4	3	2	0	1
Suzuki	1	10	2	7	0	3	1	4	0	0
Harley-Davidson	3	3	2	2	0	0	1	4	0	6
BMW	0	0	0	0	0	0	3	2	0	2
NVT	0	0	3	0	2	0	15	7	0	10

Source: NADA Motorcycle Appraisal Guide, reported in D. Purkayastha and R.D. Buzzell, "Note on the Motorcycle Industry – 1975," HBS case 578-210, p. 21.

Note: "New" NVT models in 1975 with displacements 750 cc and above are actually models that had displacements of 450-749 cc in 1968 and were modified slightly to have larger displacements.

Exhibit 6 U.K. Production of Motorcycles

Source: Motorcycle and Cycle Trader, reported in BCG, "Strategy Alternatives for the British Motorcycle Industry," p. 7-8, 18-20.

Exhibit 7 Costs Incurred to Bring a British Motorcycle to Market in the United States, 1975

Item	Dollars
Retail price	2,495
<u>Dealer markup</u>	<u>648</u>
Wholesale price	1,847
Production costs:	
Engine and frame parts	484
Assembly	121
<u>Other components</u>	<u>606</u>
Subtotal	1,211
Research and development	116
Selling and distribution costs:	
Ocean freight	60
Packing	40
Duty	75
Advertising and promotions	58
<u>Other selling and distribution</u>	<u>267</u>
Subtotal	500
Total costs	1,827
Profit contribution	20

Source: Casewriter calculations and estimates based on BCG, "Strategy Alternatives for the British Motorcycle Industry."

Note: The "engine and frame parts" line includes depreciation, raw material, and labor costs incurred in internal production of engine and frame parts. "Other components" are largely purchased from outside suppliers.

Exhibit 8 Dealer Networks in the United States, 1974/75

Competitor	Units sold (000s)	Retail value (\$ mm)	Number of dealers	Percent exclusive dealers	Units per model (000s)	Retail volume per dealer (\$000)
Honda	433.8	477.7	1,974	84	16.7	242
Yamaha	205.1	215.4	1,515	71	7.1	142
Kawasaki	129.6	147.0	1,018	61	5.6	144
Suzuki	108.5	98.1	1,103	84	4.5	89
Harley-Davidson	57.9	111.8	390	79	3.9	287
BMW	9.0	25.8	NA	NA	2.3	NA
NVT	13.9	26.3	408*	7	0.8	64*

Source: Exhibit 4 above. *Motorcycle Dealer News* Survey, R.L. Polk, and BCG estimates, reported in BCG, "Strategy Alternatives for the British Motorcycle Industry," Appendix p. 25.

* Number of Triumph dealers only. Retail volume per dealer figure assumes that all NVT volume goes through the 408 Triumph dealers.

Notes

- ¹ Boston Consulting Group, "Strategy Alternatives for the British Motorcycle Industry," London: Her Majesty's Stationery Office, 1975 (hereafter referred to as "SABMI"), p. 6.
- ² "SABMI," p. 36 and Appendix p. 204-206.
- ³ "SABMI," p. 5.
- ⁴ "SABMI," pp. 10-13.
- ⁵ D. Purkayastha and R.D. Buzzell, "Note on the Motorcycle Industry – 1975," HBS case 578-210, p. 4.
- ⁶ "SABMI," Appendix p. 9.
- ⁷ D. Purkayastha and R.D. Buzzell, "Note on the Motorcycle Industry – 1975," HBS case 578-210, p. 4.
- ⁸ "SABMI," pp. 70-73.
- ⁹ R.T. Pascale and E.T. Christiansen, "Honda (A)," HBS case 384-049 and "Honda (B)," HBS case 384-050.
- ¹⁰ "SABMI," p. 26.
- ¹¹ "SABMI," p. 28.
- ¹² "SABMI," p. 27.
- ¹³ "SABMI," pp. 28-29.
- ¹⁴ "SABMI," p. 62. The casewriter estimated the annual salary of each R&D employee by taking the monthly salary of a factory laborer given on p. 54 of "SABMI" and assuming that each R&D employee earned a salary twice that of a factory laborer.
- ¹⁵ "SABMI," p. 58.
- ¹⁶ "SABMI," pp. 59-61. The \$48 million figure comes from the discussion of strategic alternatives, "SABMI," Exhibit 28, assuming an exchange rate of 0.42 £ / \$. The ten-year depreciable life is an estimate provided by the casewriter. The 20% cost reduction associated with experience is derived from "SABMI," Exhibit 13.
- ¹⁷ "SABMI," p. 61.
- ¹⁸ "SABMI," p. 57.
- ¹⁹ "SABMI," pp. 46, 56-57.
- ²⁰ "SABMI," pp. 54, 56, 60, 62 and casewriter estimates.
- ²¹ "SABMI," Appendix p. 204.
- ²² "SABMI," pp. 63-64.
- ²³ "SABMI," pp. 58-59.
- ²⁴ D. Purkayastha and R.D. Buzzell, "Note on the Motorcycle Industry – 1975," HBS case 578-210, pp. 5-6.
- ²⁵ "SABMI," p. 30, Appendix pp. 25-29, and casewriter estimates.
- ²⁶ "SABMI," p. 30, Appendix pp. 25-29.
- ²⁷ "SABMI," p. 40.
- ²⁸ "SABMI," p. ix.